University of Central Punjab

FACULTY OF INFORMATION TECHNOLOGY

Artificial Intelligence-LAB

Term Project: Spring -2026

Term Project:

This term project aims to solidify your understanding of the basics of Machine Learning models and Neural Networks. Your task will be to implement, train, and evaluate foundational models on appropriate datasets, ensuring a hands-on grasp of their core principles and practical applications. Through this project, you will bridge theoretical knowledge with practical experience, developing essential skills in data preprocessing, model building, and performance evaluation. Collaboration, critical thinking, and creativity are key aspects of this endeavor to help you build a strong foundation in AI technologies.

Objectives and goals:

Our primary goal is to enable you to implement the basics of Machine Learning models and Neural Networks effectively. By the end of this project, you will learn:

- How to preprocess and manage datasets for machine learning tasks.
- The implementation and optimization of neural networks.
- Evaluating and interpreting model performance.
- The practical applications of machine learning in solving real-world problems.

Instructions:

Below are the detailed instructions for your term project:

1. **Team Formation:**
   - Form a team of 2-3 members.
   - Choose your teammates wisely, ensuring balanced collaboration.

2. **Project Idea Selection:**
   Select your project idea based on Machine Learning models such as:
   - Naive Bayes
   - K-Nearest Neighbors (KNN)
   - K-Means Clustering
   - Regression
   - Neural Networks

3. **Dataset Search:**
   Find a suitable dataset for your project. Recommended platforms for dataset search are given below;
   - GitHub
   - Kaggle

4. **Front-End Design:**
   - Design a user interface (front-end) for your project.
   - You may use any web-based or mobile app development language of your choice.

5. **ML Model Implementation and Integration:**
   - Implement the selected Machine Learning model using Python.
   - Ensure the model is integrated seamlessly with your front-end application.

6. **Performance Requirement:**
   - Your ML model must achieve an accuracy of 80% or higher.

7. **Result Presentation:**
   Display the results using visualizations such as:
   - Graphs
   - Confusion Matrix
   - Any other designs supported by Matplotlib or similar libraries.

8. **Submission Guidelines:**
   Submit a ZIP file containing the following:
   - Dataset used for project.
   - **.py and .ipynb** files for the code
   - A Word document explaining the project's workflow, implementation details, and results.
   - Only one person from each group will submit the project.

9. **Viva Examination:**
   - Viva will be conducted on an individual basis to assess your understanding of the project.

10. **Deadline:**
    - Submit your project by **6th June 2026.**

Evaluation:

Evaluation Criteria			
Project Submission	**Front-End Design with DB Integration**	**Viva/Presentation**	**Demo Recording**
5%	20%	60%	10%

★ *Best of Luck!* ★